CONFLUENTIA × 2UNE.AI RL HACKATHON • TRACK: SOFTWARE & DEVOPS

OpsArena: Production Cloud-Native SRE Environment

A Hardened, Deterministic Reinforcement Learning Environment for Autonomous Incident Response

8 / 8 TASKS PASSED
Pre-flight baseline checks verified in 2.77s

0.9955 LIVE SCORE
Autonomous gpt-6-luna incident rollback in 12.1s

ZERO-TRUST SANDBOX
Unprivileged UID 1000, internal-only net, no host socket

SINGLE COMMAND
docker compose up --build ... exit code forward

Author & Team: team_087 • **Evaluated Model Deployment:** gpt-6-luna (penpencil.guru) • **Demo Task:** task_02_bad_deployment

The Challenge: SRE Benchmarking for Autonomous RL Agents

Bridging the Chasm Between Synthetic Toy Problems and Fragile Sandboxes

Current Landscape Flaws	OpsArena's Engineering Solution
• **Synthetic Toy Problems:** Most benchmarks test math or simple games with zero real-world infrastructure relevance. • **Fragile Docker Sandboxes:** Real-world bash sandboxes frequently mount host sockets, leak environment credentials, or crash. • **Prompt Injection & Cheating:** Unconstrained agents rewrite test files, mock verifier functions, or leak hidden answers. • **Binary All-or-Nothing Scoring:** Agents receive 0% for making 90% diagnostic progress, discouraging nuanced exploration.	• **Production CloudMesh Simulator:** 8 interdependent microservices modeling cascading failures, pool exhaustion, and memory leaks. • **Strict Dual-Container Isolation:** Zero internet egress for sandbox, unprivileged UID 1000, no Docker socket, 512MB RAM ceiling. • **Hardened Action Allowlist:** No arbitrary bash/python exec; only 8 typed, schema-validated operational tools. • **Bounded Continuous Partial Credit:** Mathematically gated reward providing smooth optimization without breaking no-op constraints.

Dual-Container Architecture & Isolation Topology

Separation of Powers: Verifier Inaccessibility and Strict Process Containment

MODEL API LAYER	HARNESS CONTAINER (Evaluator)	SANDBOX CONTAINER (CloudMesh)
penpencil.guru (gpt-6-luna) → HTTPS Egress → model prompts & tool responses	**Dual-homed: 'internet' + 'isolated'** • Pre-flight gate orchestration • Context trimming (48k tokens) • Multi-factor programmatic verifier → HTTP /tool, /reset, /state →	**Isolated network only (internal: true)** • Unprivileged user (UID 1000) • Zero internet egress (no gateway) • 512MB RAM, 256 PIDs, cap_drop: ALL • Output strictly clipped at 4k chars

Key Security Defense	Implementation in OpsArena	Adversarial Attack Prevented
Verifier Inaccessibility	Verifier code resides strictly inside harness; sandbox exposes only /state.	Agent tampering with reward calculation or ground truth labels.
Zero Host Socket Access	/var/run/docker.sock is never mounted; cap_drop ALL; no-new-privileges.	Host escape, container breakout, and spawning rogue sibling containers.
Automated Secret Scrubbing	In-flight logs and telemetry regex-scrub database passwords, tokens, and keys.	Agent exfiltrating or leaking credentials in tool query results.

The CloudMesh Simulator: 8 Enterprise Microservices

Realistic Dependency Topologies, Dynamic Metrics, and Degradation Propagation

Service	Tier	Default Pods	Dependencies	Operational Dynamics & Incident Roles
`api-gateway`	Edge	3 pods	order, recommendation	Ingress rate limiting, circuit breaking, SSL termination, and client routing.
`order-service`	App	3 pods	payment, inventory, cache	Core transaction pipeline; subject to bad releases, canary leaks, and traffic surges.
`payment-service`	App	2 pods	db-primary, 3rd-party	Outbound payment partner integration; circuit breaker isolation under upstream latency.
`inventory-service`	App	2 pods	db-primary, db-replica	Database connection pooling; subject to connection pool starvation under checkout bursts.
`recommendation`	App	2 pods	cache-cluster	Soft dependency; fails gracefully to fallback recommendations if degraded.
`cache-cluster`	Cache	2 pods	None	Redis memory eviction storm; capacity scaling prevents thundering herd to DB.
`db-primary`	Data	1 pod	None	PostgreSQL write primary; table lock contention and migration lock timeouts.
`db-replica`	Data	2 pods	None	Read replica pool; overprovisioning causes financial drain without adding throughput.

Unified Physics: compute_service_metrics serves as the single source of truth for both model queries and verifier evaluation, guaranteeing 100% agreement between reported observations and final scoring.

Operational Action Space: 8 Typed Tools

Zero Shell Access: Structured JSON Actions with Strict Cost & Schema Bounds

Tool Name	Cost	Allowed Parameters & Types	Operational Function & Safeguards
`get_topology`	$1.00	None	Queries current service topology, pod replicas, versions, dependencies, and cumulative spend.
`query_metrics`	$2.00	service (str), metric (latency, errors, cpu, mem, rps)	Returns 5-step timeseries window with status (HEALTHY / CRITICAL). No budget leak on bad inputs.
`query_logs`	$3.00	service (str), filter (optional substring)	Structured log query with automated secret sanitization and 4,000 char bounds.
`scale_service`	$2/pod+$3	service (str), replicas (int: 1 to 20)	Scales pod pool within platform boundaries; rejects non-ints and out-of-range values.
`rollback_release`	$20.00	service (str), target_version (str)	Rolls back faulty deployments to verified stable container images in release registry.
`update_config`	$5.00	service (str), key (str), value (int/bool)	Service-specific schema: enforces int bounds (pool: 1-100, lock: 0-60k) and booleans. Rejects unauthorized keys.
`reroute_traffic`	$10.00	service (order-service), weight (int: 0 to 100)	Shifts canary traffic weights (0% to 100%) to safely isolate failing routes.
`finish`	$0.00	None	Signals completion of incident mitigation and locks state for formal verification scoring.

Budget Ceiling: Cumulative operational cost is capped at $150.00. Once the budget is exhausted, further mutating actions are rejected.

The 8-Task Incident Benchmark Suite

Seed-Diversified Scenarios Without Answer Leaks: Diagnostic Realism Across Difficulties

Task ID	Title	Diff	Simulated Root Cause & Seeded Dynamics	Target SRE Action	Oracle	No-Op
`task_01`	Redis Eviction Storm	Easy	Mem capacity (1-4GB), surge QPS (2.8k-3.6k) causes DB CPU 96%.	Scale cache-cluster ≥ 2 pods	0.996	0.000
`task_02`	Bad Deployment Rollback	Easy	Faulty release tag throws NullPointerExceptions (16-21% 500s).	Rollback order-service to v2.4.0	0.996	0.000
`task_03`	DB Pool Starvation	Med	Checkout bursts (1.6k-2.2k RPS) exhaust pool (504 timeouts).	Update db_pool_size ≥ 25	0.996	0.000
`task_04`	Slow Third-Party Gateway	Med	External payment partner hangs (6s-9s), causing thread starvation.	Enable circuit_breaker_enabled	0.996	0.000
`task_05`	Cascading Failure	Hard	Flash surge (4.4k-5.2k RPS) overwhelms pods, causing OOM crash loops.	Scale order-service ≥ 6 pods	0.994	0.000
`task_06`	Canary Memory Leak	Med	Severe memory leak on canary split route (5-15% traffic).	Reroute traffic weight to 0%	0.996	0.000
`task_07`	Overprovisioned Drain	Hard	Cluster overscaled (36 pods), burning $540/hr under 12% CPU. Safe min enforced.	Right-size cluster to ≤ 18 pods	0.996	0.000
`task_08`	Migration Lock Starvation	Hard	Exclusive table lock blocks writes (22% checkout errors, PID 4k-9k).	Set lock_timeout_ms to 1000	0.996	0.000

Programmatic Multi-Factor Reward Function

Gated Optimization: Continuous Partial Credit with Absolute No-Op Protection

Mathematical Formula: Reward = P × [0.45 × S_SLO + 0.25 × R_reg + 0.15 × B_bud + 0.15 × E_eff]

Component	Weight	Mathematical Contract & Behavior	Verification & Safeguard
P (Progress Gate)	Multiplier	P = 0.00 for no-op/idle; P ∈ (0, 0.5] for partial progress; P = 1.00 for full resolution.	Guarantees no-op agent scores 0.0000 ≤ 0.05 while rewarding partial fixes.
S_SLO (SLO Recovery)	45%	Scores post-incident error rate (≤ 0.1%) and p99 latency (≤ 100ms).	Penalizes sluggish response or incomplete error mitigation.
R_reg (Platform Stability)	25%	Evaluates non-target microservices against pre-incident baseline health.	Deducts heavily if an agent damages, downscales, or misconfigures unrelated services.
B_bud (Budget Preservation)	15%	Full marks for spend ≤ $50.00; decays linearly to 0 at the $150.00 ceiling.	Penalizes wasteful, brute-force scaling or unnecessary rollbacks.
E_eff (Action Efficiency)	15%	Full marks for resolution within ≤ 8 total model turns; decays linearly up to 40 turns.	Measures total model turns (including diagnostic reads) over aimless polling loops.

Empirical Results: Baseline Gating & Live gpt-6-luna

Verified Evidence: 8/8 Tasks Pre-Flight Pass + Autonomous Incident Resolution

1. Pre-Flight Baseline Checks (gate.json)	2. Live Model Run (gpt-6-luna on task_02)
• **Pre-flight Check Duration:** 2.77 seconds (budget: 120s) • **Oracle Solutions:** All 8 tasks score 0.994 – 0.996 (≥ 0.95 PASS) • **No-Op Agents:** All 8 tasks score exactly 0.0000 (≤ 0.05 PASS) • **Multiple Seeds Tested:** Seed 0, 42, 99 all pass with 0 errors • **Gate Enforcement:** Exit code 2 if either baseline check fails	• **Turn 1 (Triage):** Polled topology, metrics, logs; identified NPE in v2.4.2 • **Turn 2 (Remediate):** Rolled back order-service to stable v2.4.0 • **Turn 3 (Verify):** Confirmed error rate dropped from 18.4% → 0.04% • **Turn 4 (Close):** Called finish; final reward **0.9955** / 1.0000 • **Wall-Clock Elapsed: 12.1 seconds** (limit: 600s)

Sample Trajectory Evidence	Task ID	Turns	Reward	Evaluation Status & Key Finding
`trajectory_success_task_02.md`	task_02	4 turns	0.9955	SUCCESS: Clean triage → rollback → verification → sign-off in 12.1s.
`trajectory_failed_task_05.md`	task_05	1 turn (cap)	0.0000	FAILED (Negative Control): Step cap reached during read phase before scaling.

Adversarial Threat Model & Tested Security Controls

Empirically Verified Defenses Against Agent Exploitation and Cheating

Adversarial Vector	Exploit Mechanism	OpsArena Defense Mechanism	Verification Test
Task 7 Undersized Cluster Attack	Scaling all 8 services to 1 pod (8 total pods ≤ 18) to bypass cost without capacity.	Minimum safe capacity enforced per service. Under-provisioned pods trigger severe SLO breach.	`test_task_07_undersized_cluster_attack_defeated (PASS)`
Malformed Config Crash	Passing 'banana' or -500 to update_config crashing verifier.	Key-specific schema validation; rejects invalid types before state storage.	`test_malformed_config_inputs_do_not_crash_scorer (PASS)`
Output Bounds & Secret Leak	Large log dumps to overflow context or credentials leaked in tool queries.	Strict 4,000 char boundary including truncation marker; regex secret scrubber.	`test_secret_scrubbing_and_output_bounds (PASS)`
Unrelated Service Damage	Agent solves incident by wrecking other services or altering non-target configs.	Dynamic regression scoring compares non-target services to baseline health and configs.	`test_dynamic_regression_penalty (PASS)`
Budget Leak on Invalid Calls	Submitting invalid services burns budget without performing operational actions.	Arguments validated prior to deduct_cost(); invalid calls incur zero spend.	`test_cost_not_deducted_on_invalid_calls (PASS)`

Honest Threat Model: While process breakout, exploit attacks, and reward tampering are prevented by design, future iterations will expand from in-memory discrete-event simulation to multi-node Kubernetes clusters with real network packet injection.

Turnkey Submission Checklist & Reproduction

100% Contract Compliance with Single-Command Evaluation Workflow

Submission Deliverable	Required Path in team_087.zip	Compliance Status & Verification
Single-Command Compose	`docker-compose.yml`	VERIFIED: Builds sandbox on isolated net and harness on internet.
Baseline Gate Report	`baseline-results/gate.json`	VERIFIED: All 8 tasks pass (Oracle: 0.996, No-Op: 0.000) in 2.77s.
Live Model Evaluation Logs	`test-run-logs/`	VERIFIED: 3 genuine gpt-6-luna runs (run_01, run_02, run_03).
Sample Trajectories	`sample-trajectories/`	VERIFIED: 1 success trajectory (task_02) + 1 failed trajectory (task_05).
Technical Design Document	`design-doc.pdf`	VERIFIED: Exactly 2-page dense, publication-quality technical specification.
Presentation Slide Deck	`deck.pdf`	VERIFIED: Exactly 10-slide keynote presentation deck (16:9 ratio).
Comprehensive Test Suite	`tests/test_opsarena.py`	VERIFIED: 21 unit, integration, and adversarial tests passing in 7.77s.
API Key Protocol	`.env (excluded)`	VERIFIED: .env in .gitignore, excluded from team_087.zip.

One-Command Evaluation: docker compose up --build --abort-on-container-exit --exit-code-from harness

OpsArena is 100% complete, verified, and ready for competition judging. Thank you!